GG-B-814b
NOVEMBER 23, 1959
SUPERSEDING
Int. Fed. Spec. GG–B–00814a (HEW–PHS)
September 4, 1958, and
Fed. Spec. GG–B–814
December 23, 1946

FEDERAL SPECIFICATION

BURNER, ALCOHOL, PRESSURE-TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one style, grade, and size of a pressure-type alcohol burner.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

QQ–B–626 —Brass, Leaded and Non-Leaded; Rods, Shapes, Forgings and Flat Products with Finished Edges (Bars, Flat Wire and Strips).

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–P–116—Preservation, Methods of.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The brass used in the fabrication of alcohol burners shall conform to the requirements of composition 11 in Federal Specification QQ–B–626. Other materials shall be as specified in this specification.

3.2 Design.—The alcohol burners shall be of the self-generating type and of the design

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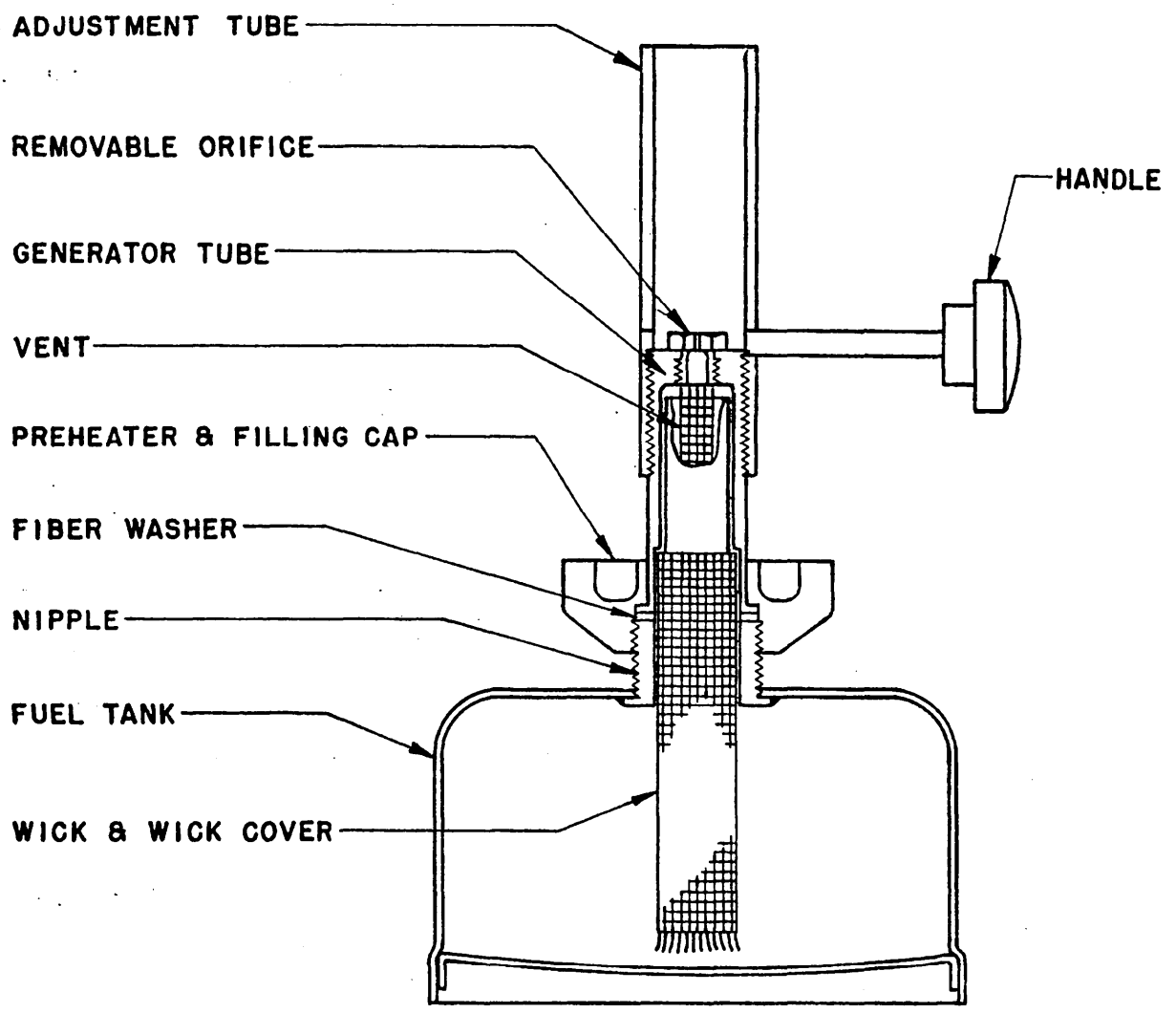


FIGURE I.

illustrated in figure 1, and shall consist essentially of a tank for fuel, a preheating and filling cap, a generator tube with flame tip, a wick and vent inside the generator tube, a burner (adjustment) tube, and a handle for adjusting the flame. The overall dimensions of the burner shall be approximately 7 inches in height and 4 inches in width.

3.3 Construction.

3.3.1 *Tank.* — The tank shall be dome-shaped, 4 inches ± ⅛ inch in diameter and 2¾ inches ± ⅛ inch in height including the flanged threaded nipple at the opening in the center of the tank top. The nipple shall be threaded to a 1 inch – 27 per inch thread and shall accept the female threaded portion of the preheating and filling cap. The flange shall be 1/16 ± 1/64 inch thick. The flanged nipple shall be 9/16 ±1/16 inch long and shall butt against the inner side of the tank opening at the top and secured thereto by brazing or silver soldering. The tank shall be made entirely of 0.032 inch thick sheet brass and shall not be reduced, at any point, to less than 0.025 inch thickness. The bottom of the tank shall be inserted into the tank body and secured

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thereto by brazing or silver soldering. The bottom edges of the tank shall be smoothly finished.

3.3.2 *Preheating and filling cap.*—The preheating and filling cap shall be 2 inches ± 1/8 inch in overall diameter, including the side fins to facilitate removal, and 3/4 ± 1/16 inch in depth. The opening in the center of the top section shall be 0.75 plus .005 minus .000 inch in diameter, to receive the generator tube, and 3/8 ± 1/32 inch in depth. A well channel, approximately 3/8 inch wide and 5/16 inch deep, shall be provided in the top section. The inner diameter of the well or channel shall be approximately 7/8 inch. The lower section of the filling cap shall be in the form of a nipple, 3/8 ± 1/16 inch long and 1 1/4 inches ± 3/32 inch in outside diameter. The center opening of the lower section shall be machined to a depth of 3/8 ± 1/32 of an inch and threaded to a 1 inch – 27 per inch thread to accept the male threaded nipple and to receive the flange of the generator tube. The preheating and filling cap shall be made of cast bronze or brass.

3.3.3 *Generator tube.*—The generator tube shall be 2 inches ± 3/32 inch long and 0.75 plus .000 minus .005 inch in outside diameter, being closed on one end with the flame tip and having a flange on the other end. The flange shall be 0.937 plus .000 minus .005 inch in outside diameter and 1/16 ± 1/32 inch in thickness. The closed end of the generator tube shall be drilled and tapped in the exact center, and threaded to take the male threaded section of the flame tip. All threads shall fit properly and be suitable for the purpose. The closed end shall be 3/16 ± 1/32 inch in thickness. The outside of the tube shall be threaded for a distance of 7/8 ± 1/16 inch from the closed end to receive the burner tube. The flanged end shall be machined to an inside diameter of 5/8 ± 1/32 inch for a distance of 1/2 ± 1/32 inch and further machined 9/16 ± 1/32 inch to a total depth of 1 7/8 inches ± 1/8 inch. The generator tube shall be made of brass. A heat- and alcohol-resistant fiber washer shall be provided with the generator tube for use between the nipple and the generator tube.

3.3.3.1 *Flame tip.*—The flame tip shall be made of brass, 3/8 ± 1/32 inch in length with one end finished to a 3/8 ± 1/64 inch flat hexagonal-shaped nut. The other end shall be 1/4 ± 1/32 inch long and threaded to a 5/16 – 27 per inch thread to accept the female threading in the closed end of the generator tube. An undercut of 1/16 ± 1/64 inch wide shall be provided between the hexagonal flat head and the threads. The threaded end of the tip shall be drilled or machined to an inside diameter of 0.187 ± .005 inch to a depth of 1/4 ± 1/64 inch. An orifice of 0.032 ± .003 inch in diameter shall be drilled in the exact center and completely through the flat hexagon head.

3.3.4 *Wick and vent.*—A standard wick of 5/8 inch in diameter, 4 3/8 inches in length, shall be bound with copper or brass wire mesh, 16 wires to the inch each way. A brass wire-mesh roll 1/4 ± 1/16 inch in diameter and 1 1/4 inches ± 1/16 inch in length, shall be used as a vent in the center of the upper section of the wick. The wire mesh shall be 12 to the inch each way. The top of the wick shall be tied with a piece of wick material to bind it to the wire mesh vent.

3.3.5 *Burner (adjustment) tube and adjusting handle.*—The burner tube shall be 3 1/4 inches ± 1/16 inch in length and 7/8 ± 1/32 inch in outside diameter. The wall of the tube shall be sufficiently thick to provide adequate rigidity. One end of the tube shall be threaded on the inside to receive the generator tube. All threads shall fit properly and be suitable for the purpose. There shall be six air vents 3/16 ± 1/32 inch in diameter, spaced equidistant around the tube 1 ± 1/16 inch from the threaded end. Two expansion slits 1 1/4 inches ± 1/16 inch in length shall be cut, one on each side of the threaded end, completely through the tubing. An adjusting handle, 1 5/8 inches ± 1/8 inch in length, shall be threaded on each end; one end secured into the burner tube 1 ± 1/8 inch from the threaded end; and the other end of the handle finished with a heat-resisting plastic knob, 5/8 ± 1/8 inch thick and 1 ± 1/8 inch in diameter. The burner tube and

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handle, except for the plastic knob, shall be made of brass.

3.4 Performance. — The burners shall be capable of vaporizing and burning alcohol with a continuous flame up to 5 inches in height, similar to that produced by blast.

3.5 Instructions.—Each burner shall be furnished with a detail instruction sheet for correct operation.

3.6 Workmanship.—Alcohol burners shall be free from defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. —Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling.—Unless otherwise specified, at least one burner shall be taken at random from each 100 or fraction thereof delivered for inspection and tests.

4.3 Inspection.—Samples shall be carefully inspected for defects in material, dimensions, workmanship, and other applicable requirements.

4.4 Tests.—Tests shall be conducted as necessary to determine conformance to the functional requirements of this specification.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Cleaning and drying.—The unit shall be cleaned and dried in accordance with the applicable requirements of Military Specification MIL–P–116, using cleaning method C–1 and any of the drying methods listed therein.

5.2 Preservation and Packaging.

5.2.1 *Level A.*—The units cleaned and dried as specified in 5.1 shall have all exterior metal surfaces coated with type P–2 preservative compound conforming to Military Specification MIL–P–116. The preserved units shall then be packaged in accordance with method IC–5 of the same specification.

5.2.2 *Level C.*—Unless otherwise specified, units cleaned and dried as specified in 5.1 shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.3 Packing.

5.3.1 *Level A.* — Twenty-four (24) units preserved and packaged as specified in 5.2.1 shall be packed in a class 2, style 2, 2½, 4 or 4½ nailed wood box conforming to Federal Specification PPP–B–621. The box shall be closed in accordance with the applicable requirements of Federal Specification PPP–B–621, with strapping to be in accordance with the appendix thereto, using galvanized strapping.

5.3.2 *Level B.* — Twenty-four (24) units preserved and packaged as specified in 5.2.1 shall be packed in the same manner as specified in 5.3.1, except that a class 1, style 2, 2½, 4 or 4½ nailed wood box and coated-finish strapping shall be used.

5.3.3 *Level C.*—The units shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated

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Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.4 Marking.

5.4.1 *Civil agencies.* — In addition to any special marking required in the contract or order, marking shall be in accordance with Federal Standard No. 123.

5.4.2 *Military agencies.*—In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Sampling required (see 4.2).

(*c*) Selection of the applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item for which a Federal Stock Number is available:

6640–410–2820—Burner, Alcohol, Barthel Type, 8 oz. Capacity.

6.3 Transportation description.—Transportation description applicable to this item is:

Burners, Lamp.
Carload minimum weight 16,000 pounds, subject to Rule 34, Uniform Freight Classification.

Burners, Lamp.
Motor volume minimum weight 16,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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